

Bright Yellow 'Golden Tree' Coral Is So Strange It Needed a New Family

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Deep in the Pacific Ocean off Costa Rica, scientists have found a bright yellow coral so unusual that it does not fit into any known coral family. To classify it, they had to create a brand-new family.

The coral grows in branching shapes that look like trees. Some **colonies** are more than a meter tall, which is over 3 feet. It was found between 360 and 529 meters below the surface, on underwater mountains called **seamounts**. Researchers used a remotely operated underwater robot named SuBastian to collect samples during expeditions in 2019 and 2023.

The coral grew on rocky spots where **brittle stars**, relatives of starfish, covered almost every surface. The yellow branches seemed to rise from the seafloor like glowing trees surrounded by stars.

That scene inspired its name. The scientists used Quenya, the Elvish language invented by J.R.R. Tolkien, author of "The Lord of the Rings." The group name Laurinque means "golden tree," and the **species** name elenya means "of the stars."

Figuring out where the coral belonged was tricky. Scientists first checked three **genes** often used to sort corals, but each gene pointed to a different family. So they compared hundreds of **genes** at once. That showed the coral is a separate branch on the **family tree**, most closely related to a family called Eunicellidae but different enough to get its own family, Laurinqueidae.

Lead researcher Odalisca Breedy of the University of Costa Rica said she was amazed by the beauty of the coral garden when she first saw it.

The discovery shows how much of the deep ocean is still unexplored. Coral gardens like this give homes to many other deep-sea creatures. The study was published in the journal ZooKeys.

Word Treasure

seamounts -- Mountains that rise from the ocean floor but stay underwater.

colonies -- Groups of tiny coral animals living joined together as one structure.

brittle stars -- Sea animals related to starfish, with long, thin, bendy arms.

genes -- Instructions inside cells that help decide how a living thing grows and what it is like.

species -- A group of living things that are alike and can have young together.

family tree -- A chart showing how living things are related to one another.

Background you need

Corals are animals made of many tiny creatures living together, and some grow in the deep, dark ocean far from sunlight.

Scientists sort living things into groups such as species, genus and family based on how closely related they are.

Comparing genes lets scientists see which living things share common ancestors, which helps them place a new find on the family tree.

Much of the deep ocean has never been explored, so researchers use underwater robots to study and collect creatures there.

Break it down

LEAD: A bright yellow coral off Costa Rica fits no known coral family

KEY FINDING: Scientists create a new family, Laurinqueidae

+ - WHERE: Seamounts 360 to 529 meters deep in the Pacific

+ - HOW: Collected by the underwater robot SuBastian in 2019 and 2023

+ - NAME: Elvish words meaning golden tree and of the stars

EVIDENCE: Three genes disagreed, so hundreds of genes were compared

WHY IT MATTERS: Much of the deep ocean is unexplored; coral gardens shelter other life

Why it matters

The find shows how much of the deep ocean remains unknown and why coral gardens, which give homes to many deep-sea creatures, are worth studying and protecting.

Test what you remember

1. Near which country was the coral found?

- ☐ (A) Costa Rica
- ☐ (B) Japan
- ☐ (C) Mexico
- ☐ (D) Australia

2. What color is the new coral?

- ☐ (A) Deep red
- ☐ (B) Bright green
- ☐ (C) Pale blue
- ☐ (D) Bright yellow

3. What does the name Laurinque mean?

- ☐ (A) Yellow flower
- ☐ (B) Deep ocean
- ☐ (C) Of the stars
- ☐ (D) Golden tree

4. What language inspired the coral's name?

- ☐ (A) Quenya, an Elvish language
- ☐ (B) Spanish
- ☐ (C) Greek
- ☐ (D) Latin

5. Why did scientists compare hundreds of genes?

- ☐ (A) To count its branches
- ☐ (B) They wanted to change its color
- ☐ (C) To find out its age
- ☐ (D) Three genes pointed to different families

6. What tool did researchers use to collect samples?

- ☐ (A) Scuba divers
- ☐ (B) A submarine named Falkor
- ☐ (C) A fishing net
- ☐ (D) A remotely operated underwater robot named SuBastian

Answer key (try the quiz first!): 1: A 2: D 3: D 4: A 5: D 6: D

1. Lead researcher Odalisca Breedy: Says she was amazed by the beauty of the coral garden when she first saw it.